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Teorema de Pitágoras

Teorema 1 (de Pitágoras). *Sea H un espacio prehilbert y $\{x_i\}_{i=1}^n$ un sistema ortogonal*

$$\implies \left\| \sum_{i=1}^n x_i \right\|^2 = \sum_{i=1}^n \|x_i\|^2 .$$

***Demostración*:** Por defінción de norma inducida, tenemos que

$$\left\| \sum_{i=1}^n x_i \right\|^2 = \left\langle \sum_{i=1}^n x_i, \sum_{j=1}^n x_j \right\rangle = \sum_{i=1}^n \sum_{j=1}^n \langle x_i, x_j \rangle .$$

Por ortogonalidad, $\langle x_i, x_j \rangle = 0$ si $i \neq j$ y $\langle x_i, x_i \rangle = \|x_i\|^2$, luego $\left\| \sum_{i=1}^n x_i \right\|^2 = \sum_{i=1}^n \|x_i\|^2$. ∎

Referenciado en

- `teo-riesz-fischer`
- `prop-sistema-ortogonal-indep-lineal`
- `teo-identidad-plancherel`

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